

Digital Design

RTL Design

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ASM/FSMD or RTL Based Design

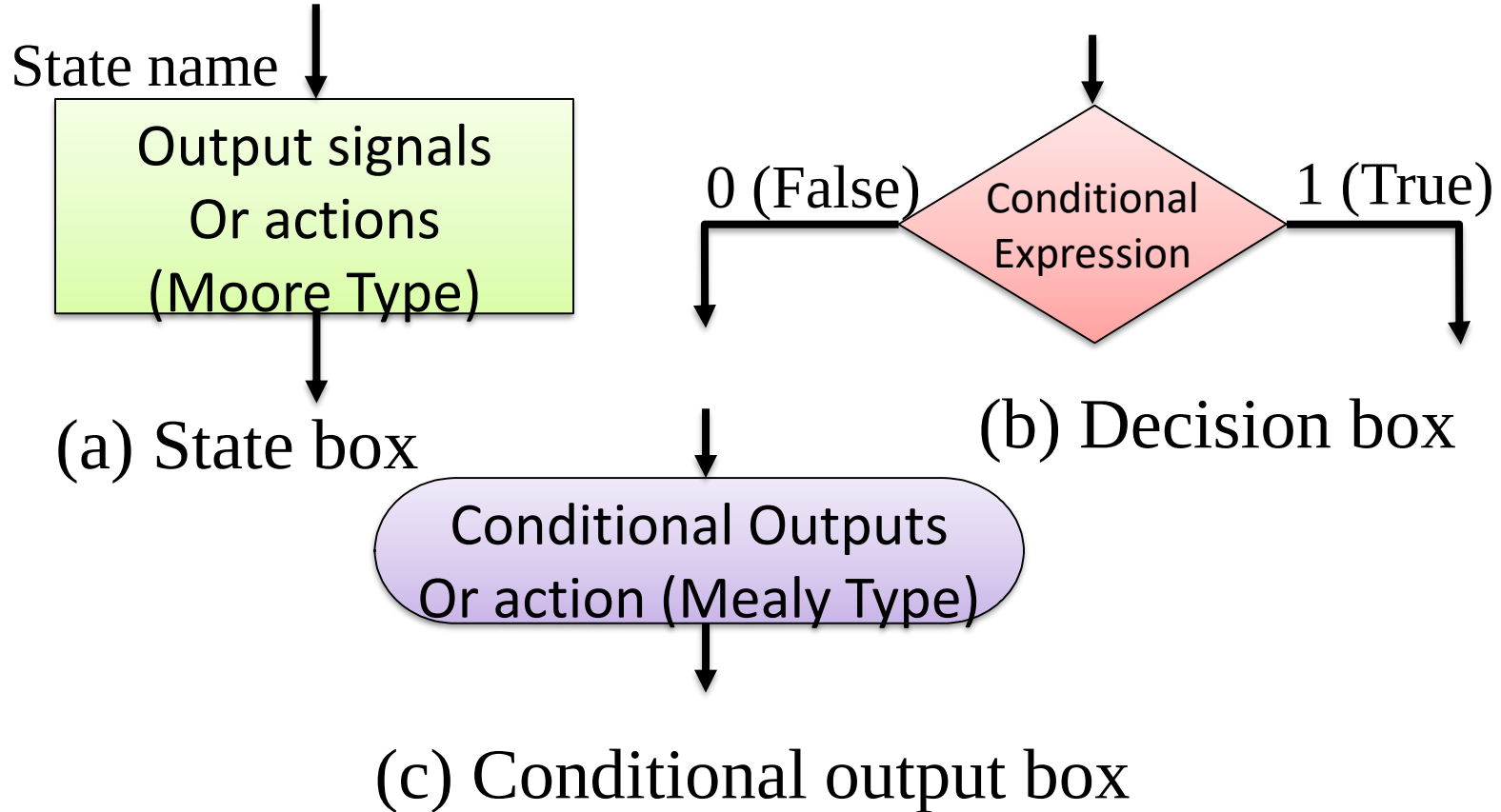
RTL Design

- Drawbacks of FSM
- $ASM = FSM + DP$
- ASM building blocks
- Modeling FSM in ASM

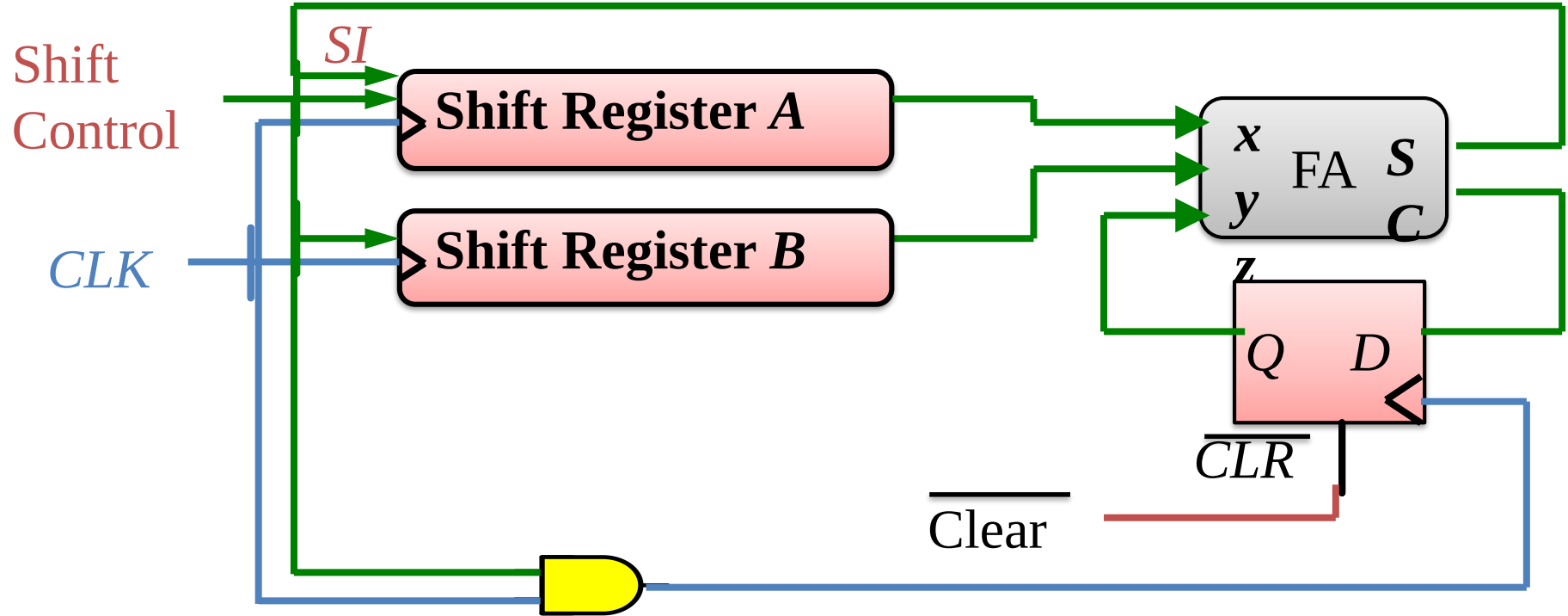
Reference Material

- Chapter 8 of Mano Book 6th Edition
 - Design at Register Transfer Level
 - Classic Example: Booth Multiplication
- Chapter 15 of Kumar Book
 - A. Anand Kumar, *Fundamentals of Digital Circuits* 3rd Edition, PHI. 2014

ASM charts : 3 Elements used

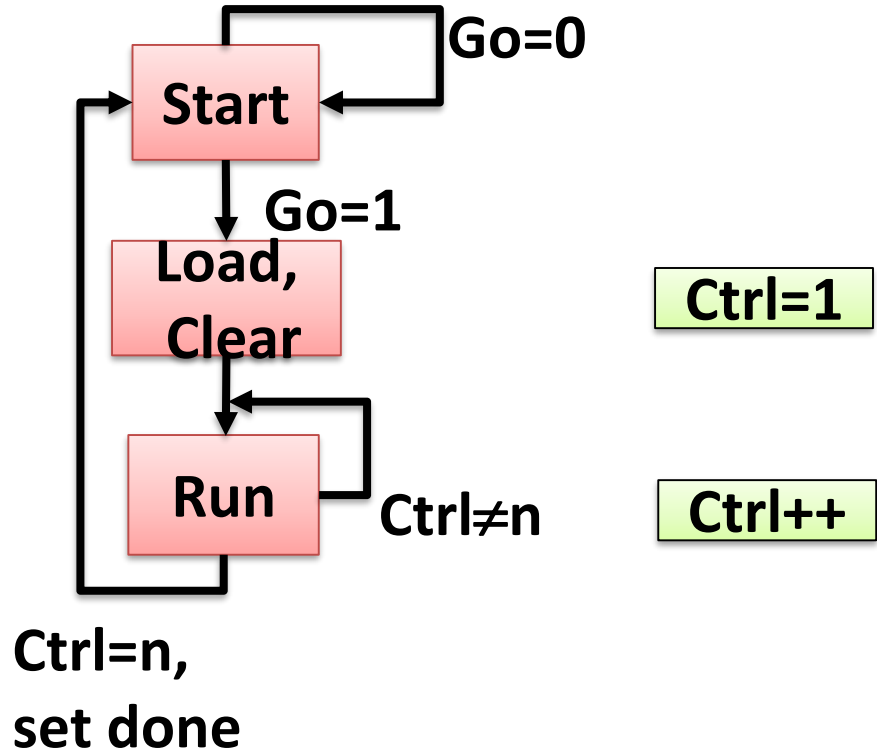


Remember : Serial Addition : EX 7



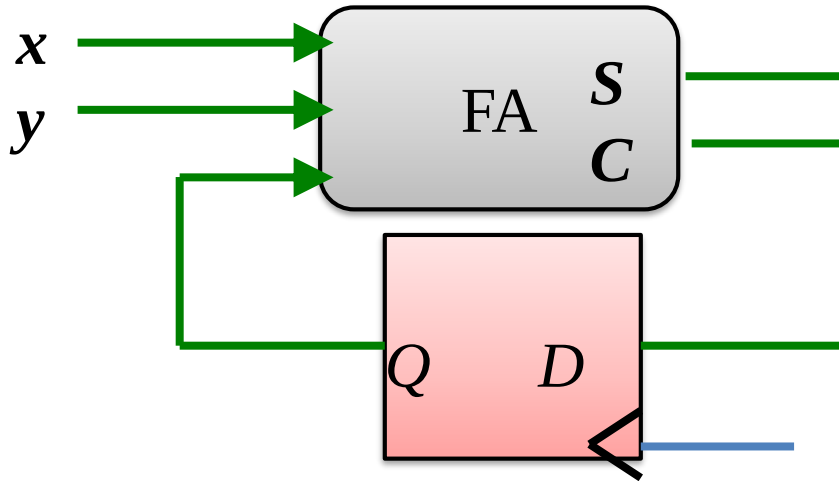
ASM Overview : High Level

- Ex. Serial Addition
- Control Part/Path

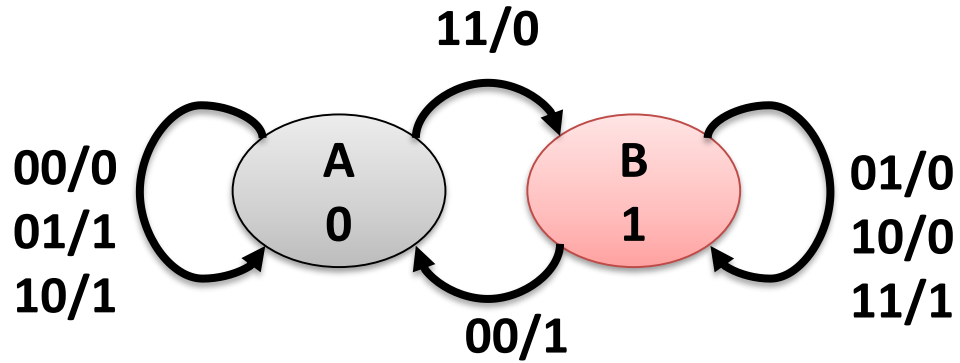


Remember : Serial Addition: EX 7

- Model S in terms of X, Y and Q (State)

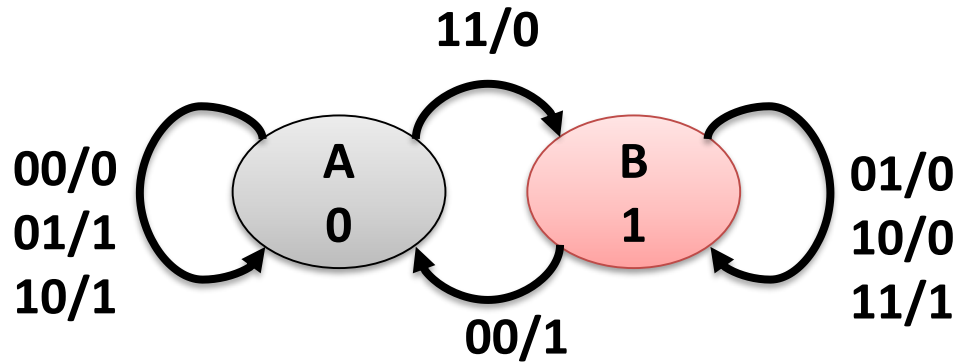


Mealy FSM for Binary Adder

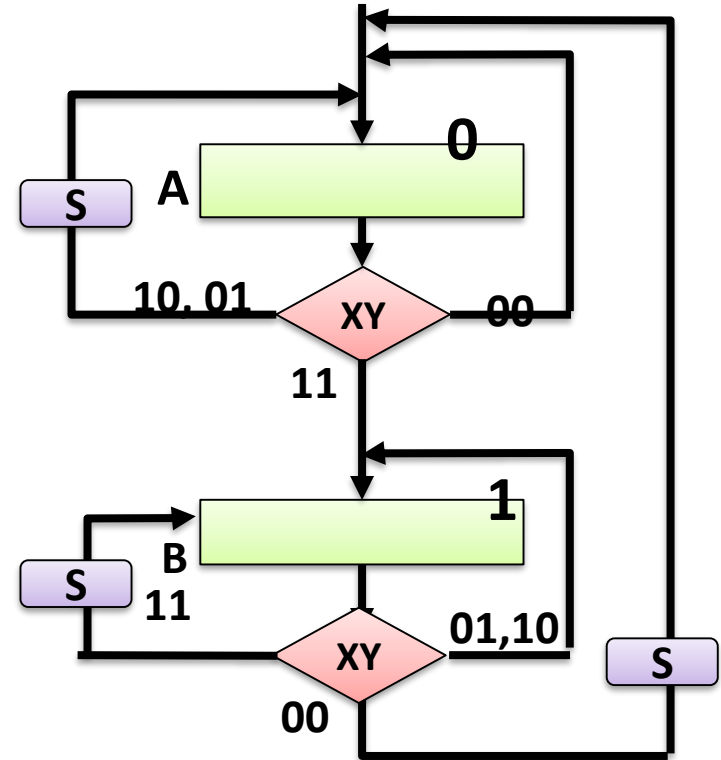


FSM

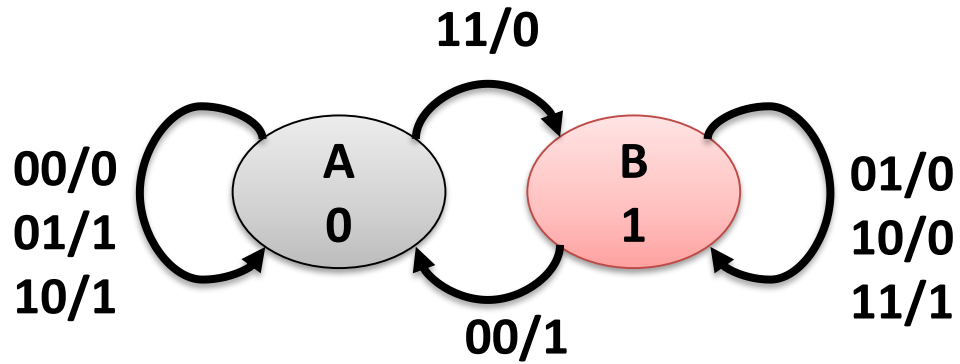
Mealy FSM for Binary Adder: Ex 7



FSM

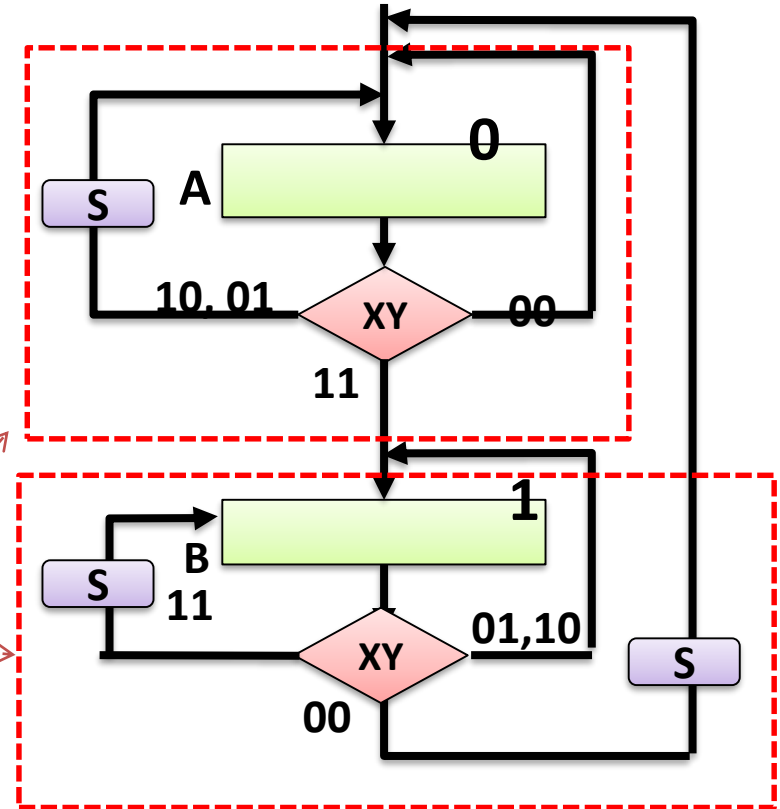


Mealy FSM for Binary Adder



FSM

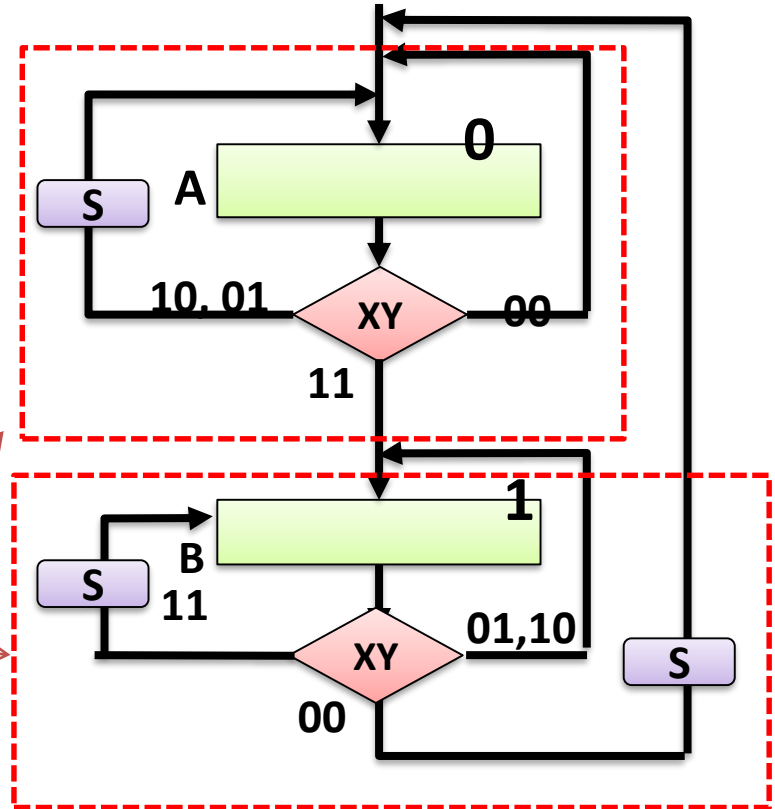
ASM
Blocks



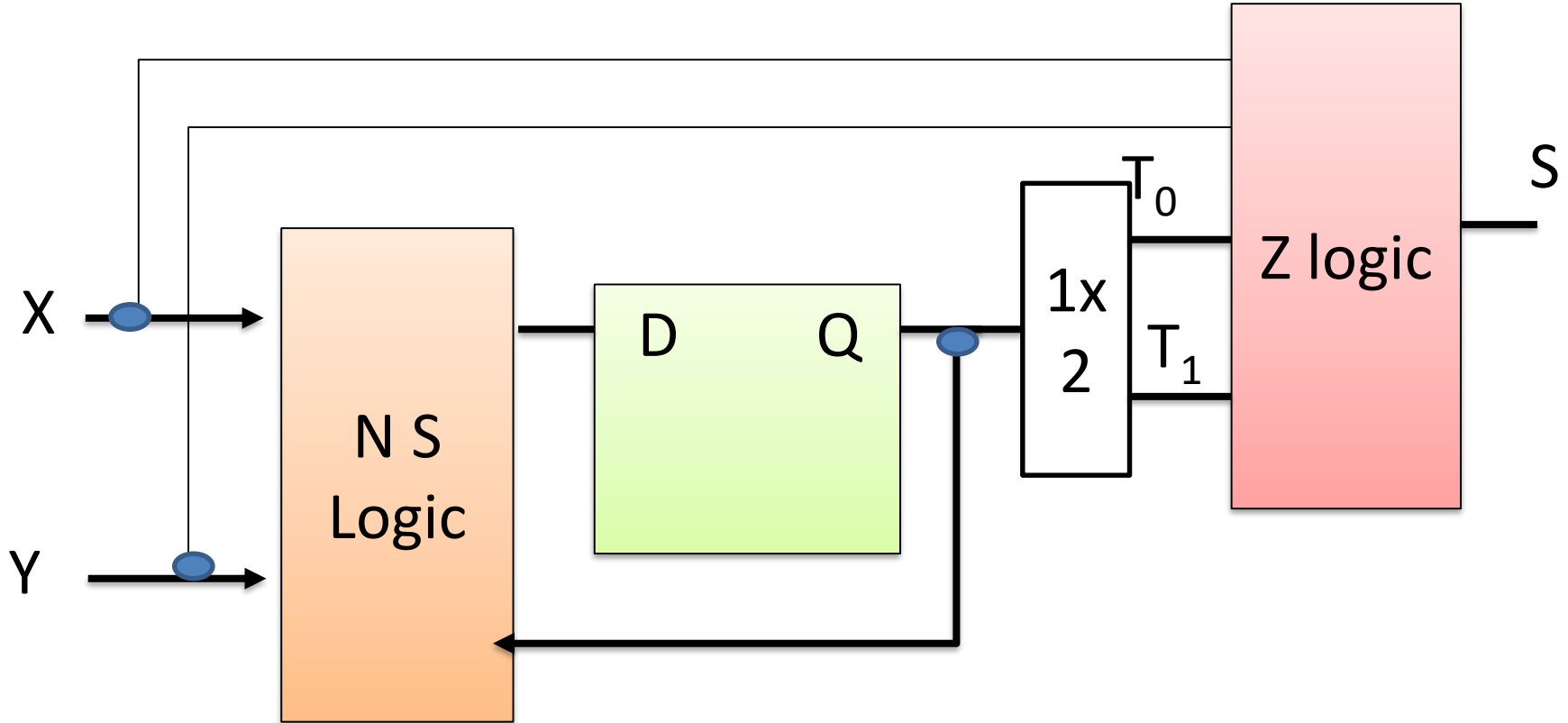
Mealy FSM for Binary Adder

- ASM Blocks
 - Two blocks in this example
- An ASM Block
 - Include a state and all its outgoing edges, condition boxes and conditional state boxes
 - All the parts of an ASM block execute in one cycle

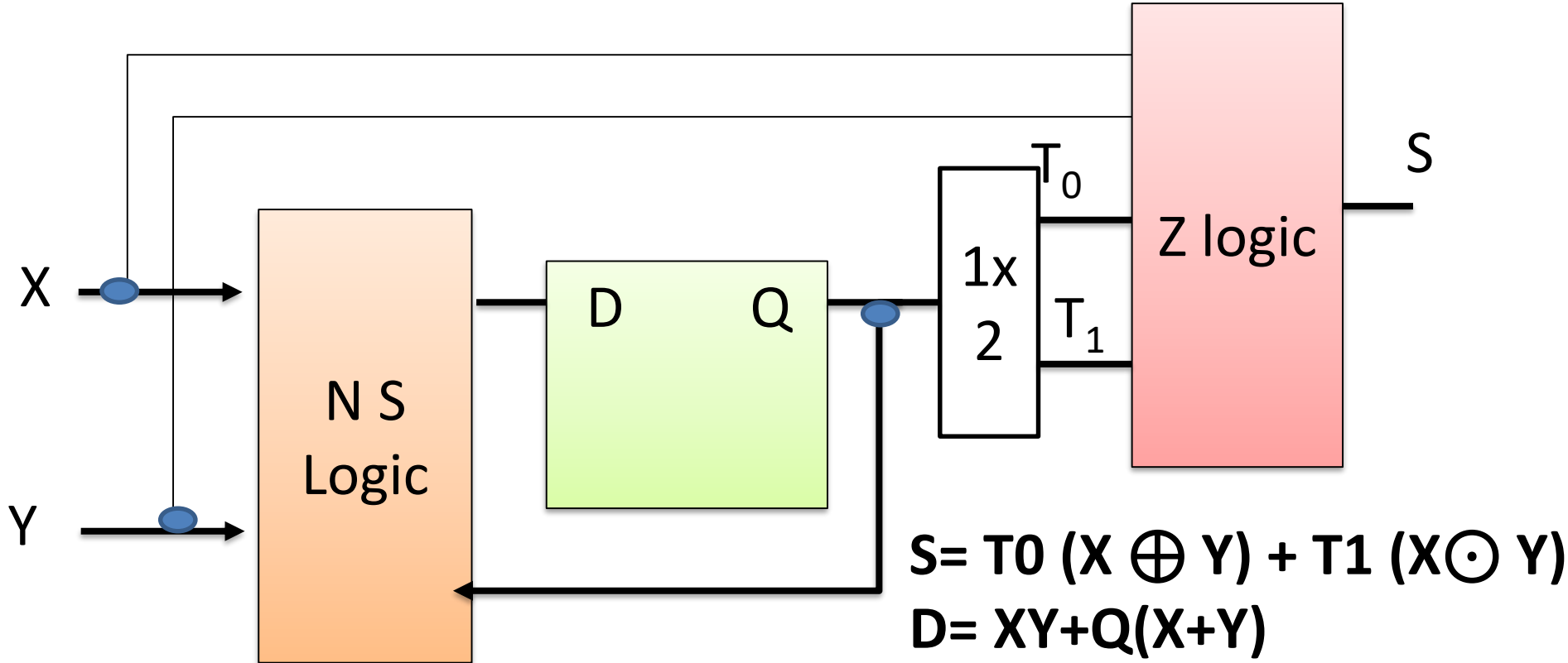
ASM
Blocks



Mealy FSM for Binary Adder



Mealy FSM for Binary Adder

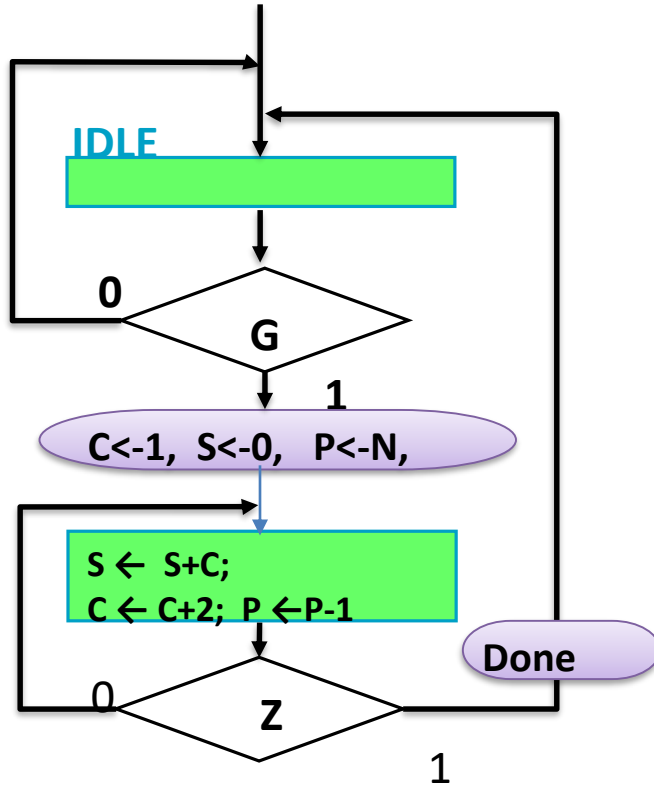


Example 8:

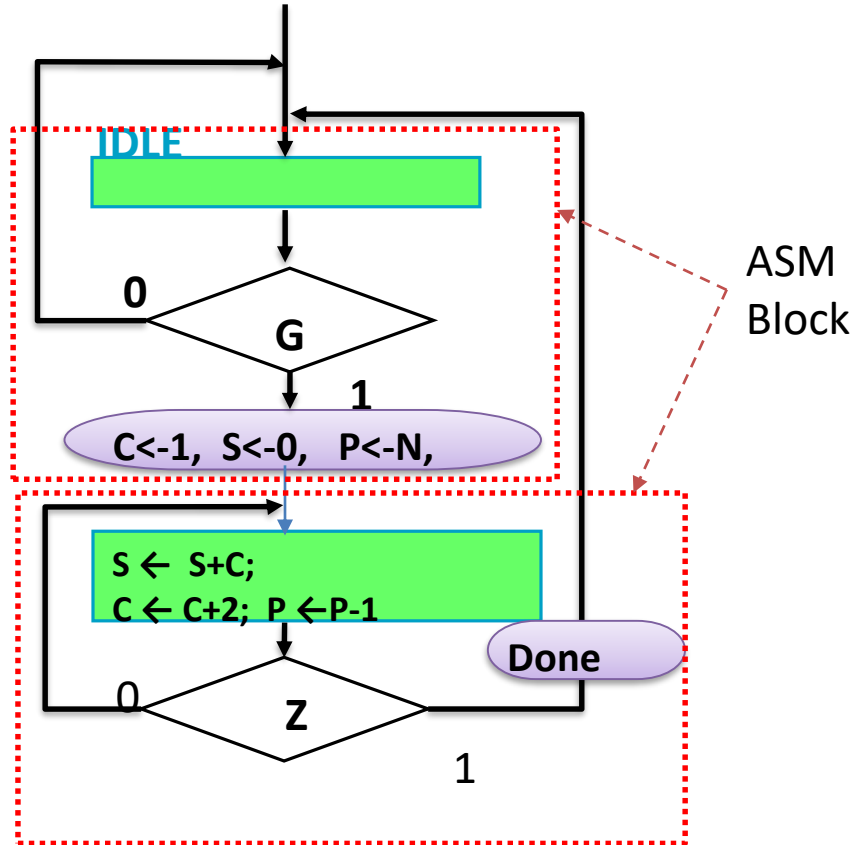
Design a Digital Circuit to Sum of First N odd numbers

It is N^2 but we want to use basic summation method $1+3+5+7..+(2n-1)$

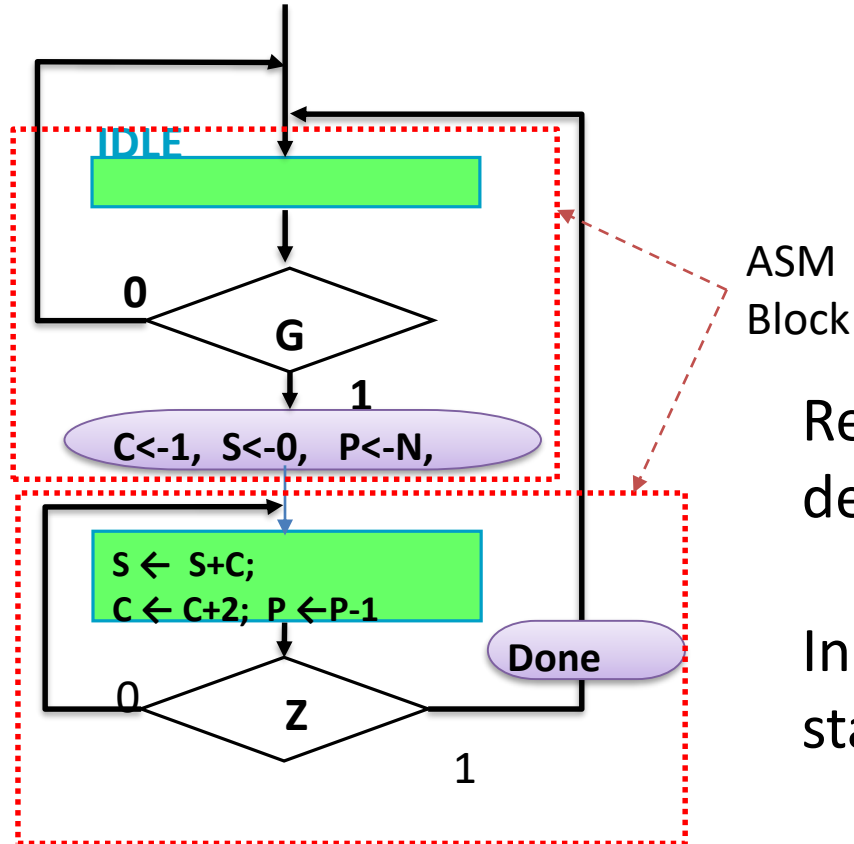
Sum of First N odd numbers



Sum of First N odd numbers



Sum of First N odd numbers



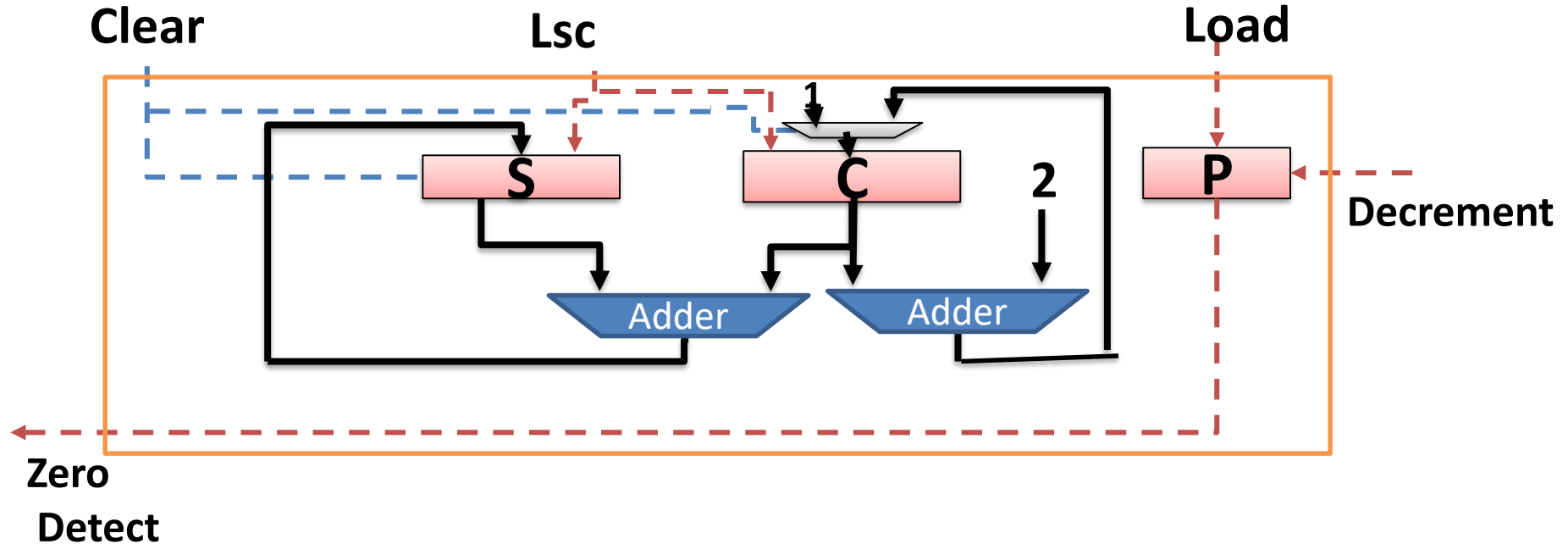
Require : two adders,
decrement register, S reg, C reg

In state : We can put RTL like
statement

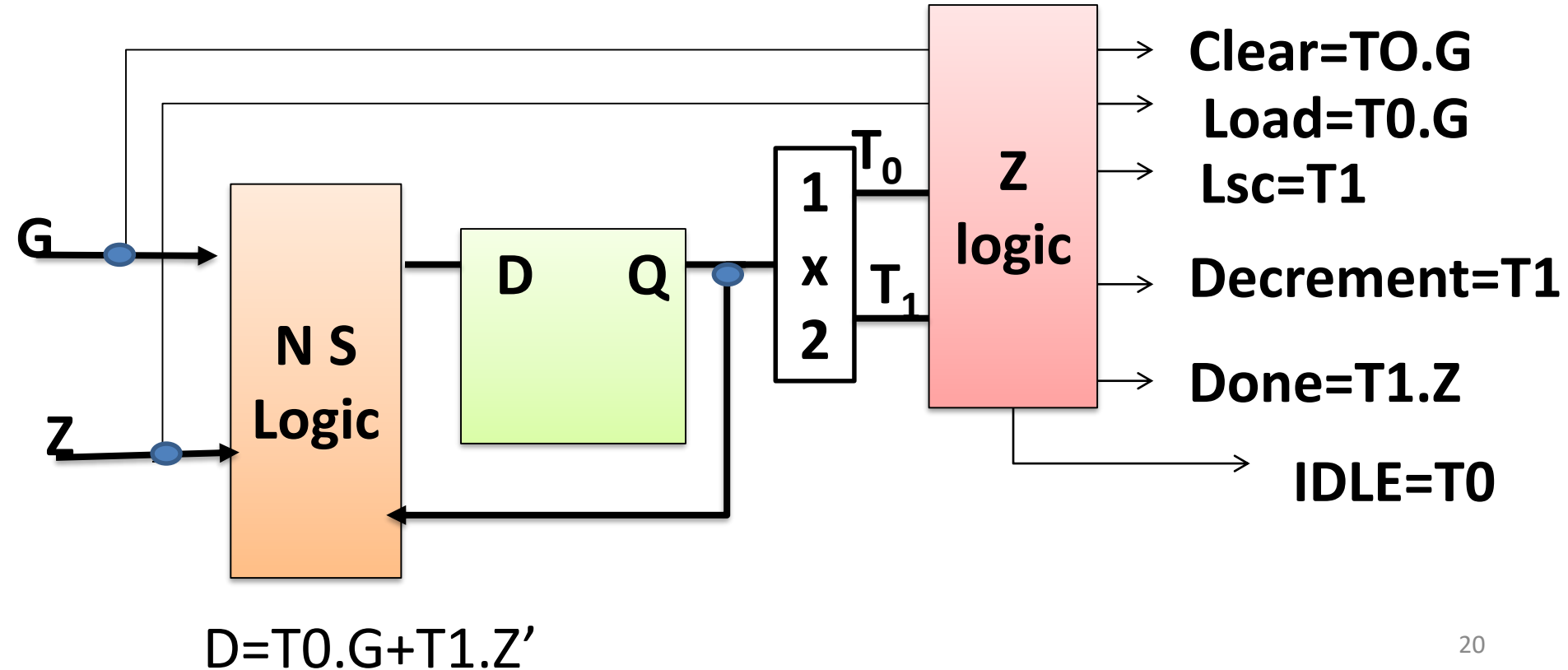
$$S \leftarrow S + C, C \leftarrow C + 2, P \leftarrow P - 1$$

$$C \leftarrow -1, S \leftarrow 0, P \leftarrow N$$

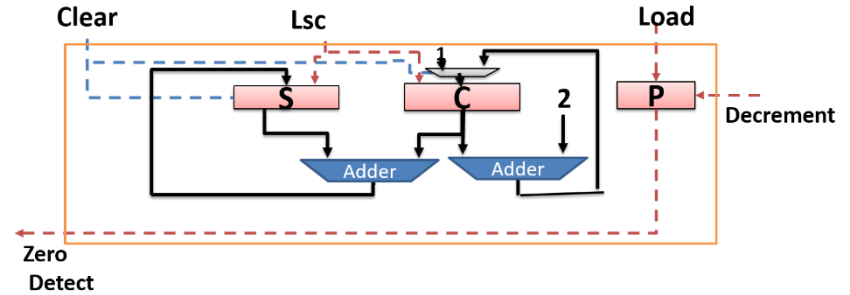
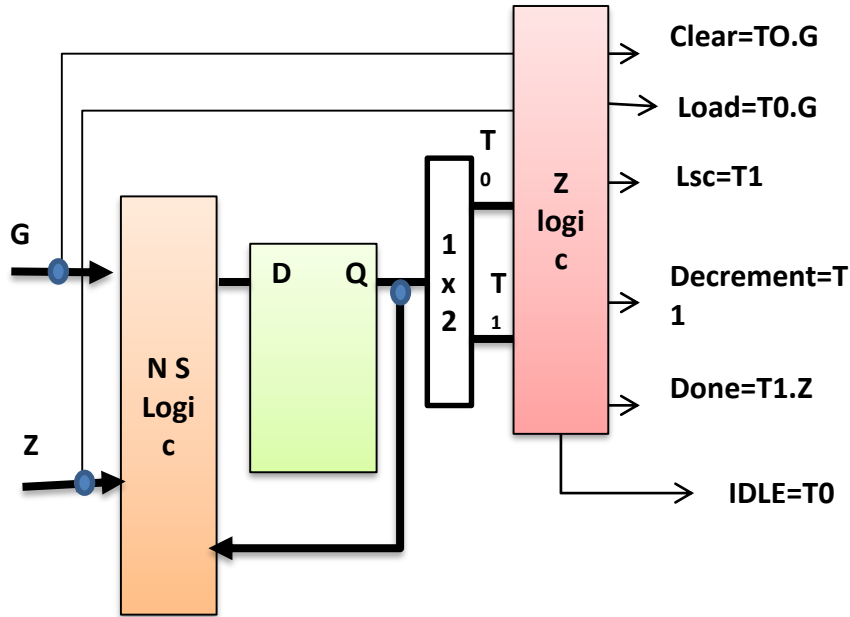
DP: Sum of First N odd numbers



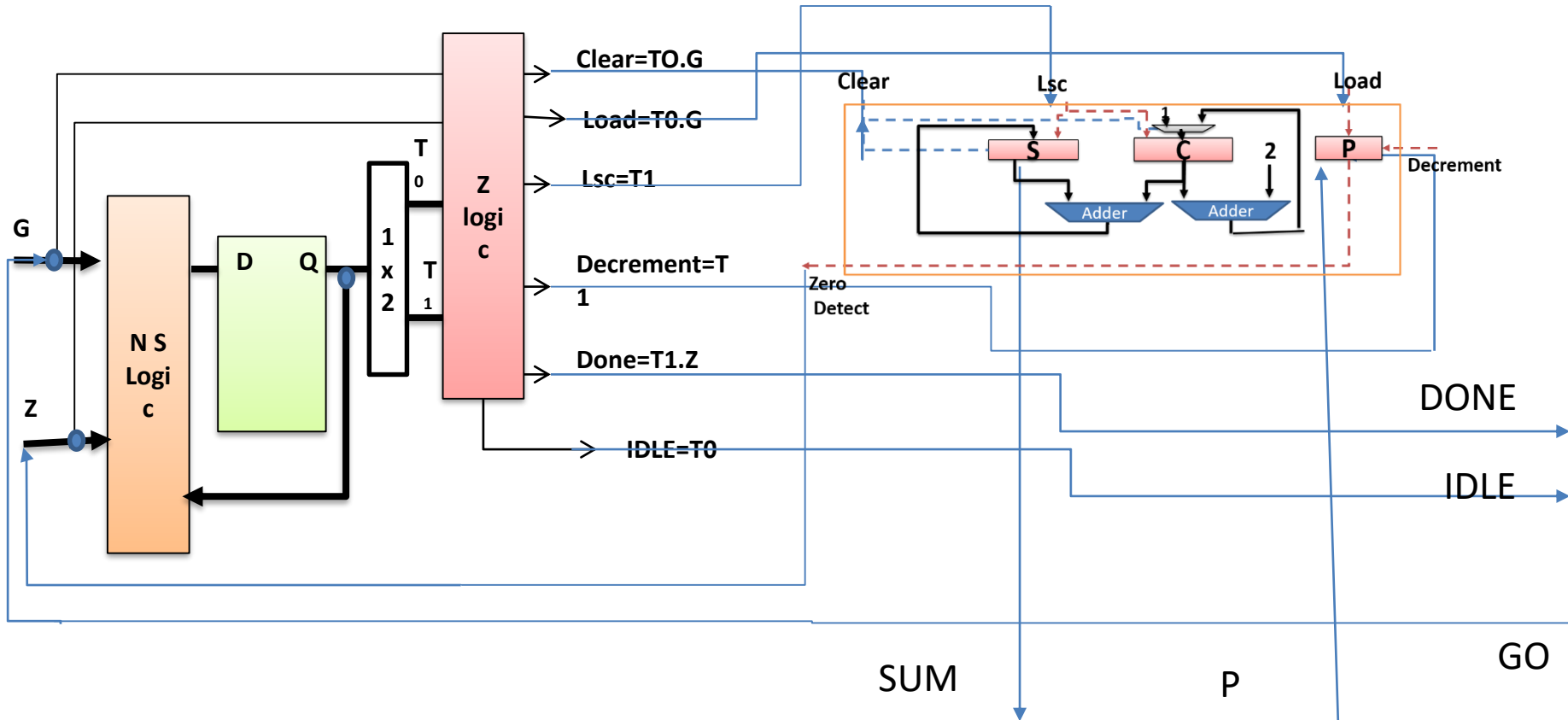
CP: Sum of First N odd numbers



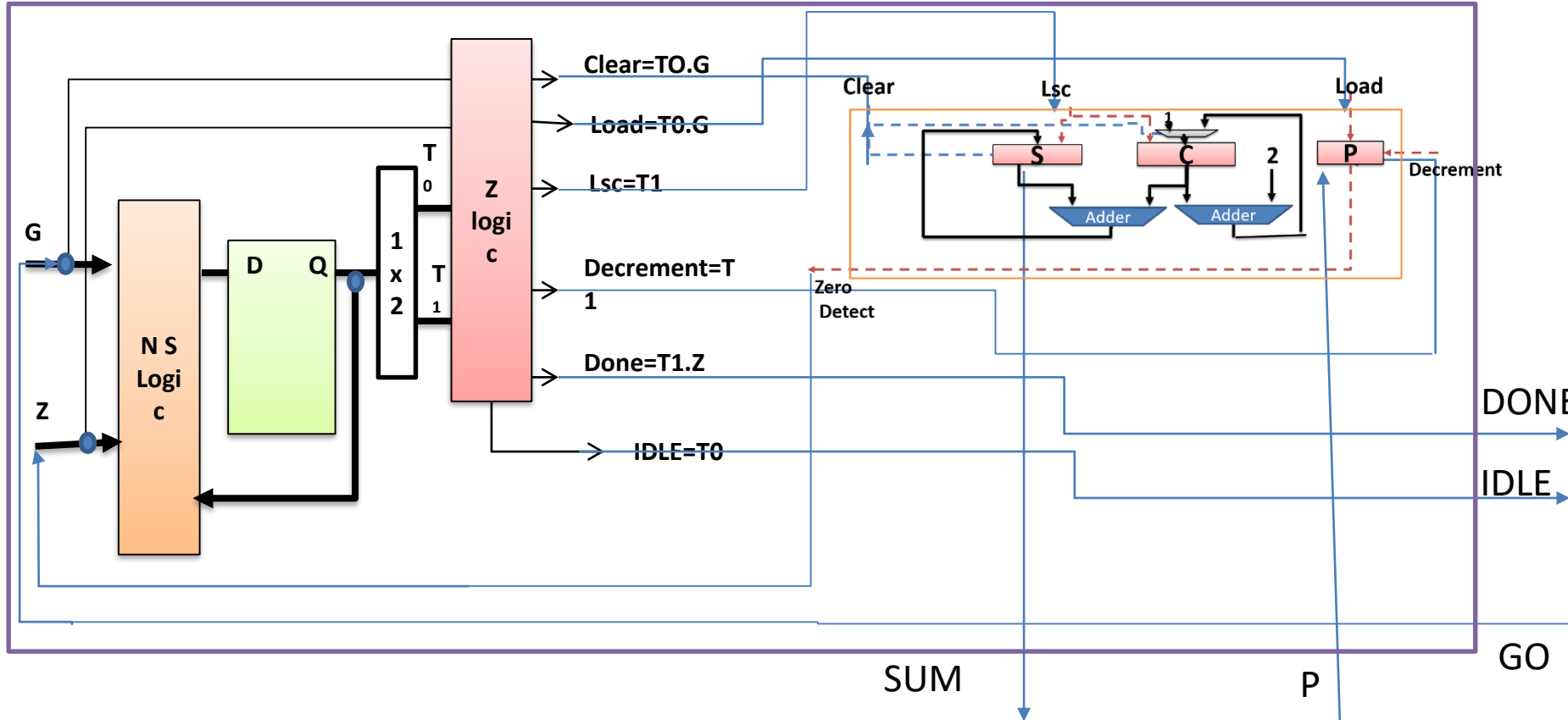
CP+DP: Sum of First N odd numbers



CP+DP: Sum of First N odd numbers



CP+DP: Sum of First N odd numbers



CP+DP: Sum of First N odd numbers

Black Box Unit
For
Sum of First N Odd Number

SUM

P

DONE

IDLE

GO